

Excess Splitting for Inf-Convolutions of Multiple ΛVaR^+ Risk Measures

A corrected progress note

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Abstract

This note revisits the open problem of characterizing the inf-convolution of several ΛVaR^+ risk measures. The main result is an exact level-wise reduction: for candidate capital levels $m_1, \dots, m_n$, with $s = \sum_i m_i$, feasibility is equivalent to being able to split the positive excess $(X - s)_+$ into nonnegative pieces Z_i satisfying the translated tail constraints

$$\mathbb{P}(Z_i > r) \leq \Lambda_i(m_i + r), \quad r > 0.$$

This gives a rigorous formula for the value and a rigorous attainment and allocation criterion. I also give a corrected Kantorovich–Hall criterion for a relaxed, randomized version of the splitting problem; it is exact on rich extensions carrying an independent randomizer, but it is not equivalent to actual splitting on an arbitrary atomless space. A counterexample is included. Thus the note gives genuine partial progress and an exact reformulation, but not a closed-form solution of Liu’s general multiple- ΛVaR^+ problem.

1 Status of the result

The problem considered here is the one identified in Liu’s paper on risk sharing with Lambda Value-at-Risk and in the associated open-problem entry: characterize

$$\left(\square_{i=1}^n \Lambda_i \text{VaR}^+ \right)(X)$$

and the existence and structure of optimal allocations. The result below is best read as follows.

- Theorem 3.2 is exact. It gives a necessary and sufficient condition for a vector of capital levels $m = (m_1, \dots, m_n)$ to be achievable.
- Theorem 4.1 gives a corresponding exact attainment criterion and constructs optimal allocations whenever the reduced splitting problem attains its minimum.
- Theorem 5.1 is a rigorous replacement for the informal Kantorovich-dual argument in the earlier draft. It is exact for a relaxed randomized problem and becomes exact for the original problem on a probability space rich enough to carry an independent randomizer.
- Proposition 5.4 shows that this richness assumption is not cosmetic: on a standard atomless space the relaxed Kantorovich condition can hold while no actual split exists.

Consequently, this note should not be advertised as a closed-form solution of the open problem. It is a rigorous exact reduction plus a rigorous relaxed LP / dual criterion, hence substantive partial progress.

2 Setup and endpoint conventions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{X}$ be a solid linear space of real-valued random variables containing the constants; for instance one may take L^p , $1 \leq p \leq \infty$, or a solid ideal with $L^\infty \subseteq \mathcal{X} \subseteq L^1$. Solidity means that if $Y \in \mathcal{X}$ and $|Z| \leq |Y|$ a.s., then $Z \in \mathcal{X}$.

For a random variable Y , write

$$F_Y(x) = \mathbb{P}(Y \leq x), \quad \bar{F}_Y(x) = \mathbb{P}(Y > x).$$

Let $\Lambda_i : \mathbb{R} \rightarrow [0, 1]$ be right-continuous and not identically equal to 1. Define

$$\rho_i(Y) = \Lambda_i \text{VaR}^+(Y) := \sup\{x \in \mathbb{R} : F_Y(x) < 1 - \Lambda_i(x)\} = \sup\{x \in \mathbb{R} : \mathbb{P}(Y > x) > \Lambda_i(x)\}, \quad (1)$$

with $\sup \emptyset = -\infty$ and with value $+\infty$ if the set is unbounded above.

Because ρ_i may take the value $-\infty$, it is cleaner to state the risk-sharing value in upper-level form:

$$\mathcal{V}(X) := \inf \left\{ \sum_{i=1}^n m_i : \begin{array}{l} m_i \in \mathbb{R}, X_i \in \mathcal{X}, \sum_{i=1}^n X_i = X \text{ a.s.,} \\ \rho_i(X_i) \leq m_i \text{ for every } i \end{array} \right\}. \quad (2)$$

When all relevant risk values are finite real numbers, this is exactly the usual inf-convolution $\inf \sum_i \rho_i(X_i)$. More generally, it is the usual inf-convolution with the natural upper-sum convention that avoids the ambiguous expression $+\infty - \infty$.

Lemma 2.1 (Acceptance at a deterministic level). *For $m \in \mathbb{R}$ and $Y \in \mathcal{X}$,*

$$\rho_i(Y) \leq m \iff \mathbb{P}(Y > t) \leq \Lambda_i(t) \text{ for every } t > m. \quad (3)$$

Equivalently,

$$\rho_i(Y) \leq m \iff \mathbb{P}((Y - m)_+ > r) \leq \Lambda_i(m + r) \text{ for every } r > 0. \quad (4)$$

Proof. Let

$$B_i(Y) := \{t \in \mathbb{R} : \mathbb{P}(Y > t) > \Lambda_i(t)\}.$$

By definition, $\rho_i(Y) = \sup B_i(Y)$. The inequality $\sup B_i(Y) \leq m$ is equivalent to saying that no $t > m$ belongs to $B_i(Y)$, which is exactly (3). Substituting $t = m + r$ gives (4). $\square$

3 The exact excess-splitting reduction

Fix $X \in \mathcal{X}$. For $m = (m_1, \dots, m_n) \in \mathbb{R}^n$, define

$$s(m) := \sum_{i=1}^n m_i, \quad W_m := (X - s(m))_+,$$

and the translated tail budgets

$$b_i^m(r) := \Lambda_i(m_i + r), \quad r > 0. \quad (5)$$

Since $\mathcal{X}$ is solid and contains constants, $W_m \in \mathcal{X}$.

Definition 3.1 (Actual tail-capacitated excess split). *Let $\mathcal{Z}_m(X)$ be the set of all n -tuples $(Z_1, \dots, Z_n)$ of nonnegative random variables in $\mathcal{X}$ such that*

$$\sum_{i=1}^n Z_i = W_m \quad \text{a.s.}, \quad (6)$$

$$\mathbb{P}(Z_i > r) \leq b_i^m(r) = \Lambda_i(m_i + r) \quad \text{for all } i = 1, \dots, n \text{ and all } r > 0. \quad (7)$$

We say that m is excess-feasible for X if $\mathcal{Z}_m(X) \neq \emptyset$.

Theorem 3.2 (Exact excess-splitting formula). *For every $X \in \mathcal{X}$,*

$$\mathcal{V}(X) = \inf \left\{ \sum_{i=1}^n m_i : m \in \mathbb{R}^n, \mathcal{Z}_m(X) \neq \emptyset \right\}. \quad (8)$$

Thus a total level $s = \sum_i m_i$ is achievable precisely when the positive excess $(X - s)_+$ can be split into pieces whose tails fit the translated Λ_i -budgets.

Proof. First suppose that $m \in \mathbb{R}^n$ and $(Z_1, \dots, Z_n) \in \mathcal{Z}_m(X)$. Put $s = s(m)$ and $D = (s - X)_+$. Choose any nonnegative $D_1, \dots, D_n \in \mathcal{X}$ with $\sum_i D_i = D$; for example, take $D_1 = D$ and $D_i = 0$ for $i \geq 2$. Define

$$X_i := m_i + Z_i - D_i, \quad i = 1, \dots, n. \quad (9)$$

Then

$$\sum_{i=1}^n X_i = s + \sum_i Z_i - \sum_i D_i = s + (X - s)_+ - (s - X)_+ = X.$$

Moreover $X_i \leq m_i + Z_i$, and hence, for every $r > 0$,

$$\mathbb{P}(X_i > m_i + r) \leq \mathbb{P}(Z_i > r) \leq \Lambda_i(m_i + r).$$

By Lemma 2.1, $\rho_i(X_i) \leq m_i$ for every i . Hence $\mathcal{V}(X) \leq \sum_i m_i$. Taking the infimum over all excess-feasible m gives one inequality in (8).

Conversely, suppose $X_1, \dots, X_n \in \mathcal{X}$ and finite levels $m_1, \dots, m_n \in \mathbb{R}$ satisfy

$$\sum_i X_i = X, \quad \rho_i(X_i) \leq m_i \quad \text{for every } i.$$

Let $s = \sum_i m_i$, $W = (X - s)_+$, and

$$E_i := (X_i - m_i)_+.$$

Lemma 2.1 gives

$$\mathbb{P}(E_i > r) = \mathbb{P}(X_i > m_i + r) \leq \Lambda_i(m_i + r), \quad r > 0. \quad (10)$$

Pointwise,

$$W = \left(\sum_{i=1}^n (X_i - m_i) \right)_+ \leq \sum_{i=1}^n (X_i - m_i)_+ = \sum_{i=1}^n E_i.$$

Define, on the set $\{\sum_j E_j > 0\}$,

$$Z_i := W \frac{E_i}{\sum_{j=1}^n E_j},$$

and set $Z_i = 0$ on $\{\sum_j E_j = 0\}$. Then $0 \leq Z_i \leq E_i$, $\sum_i Z_i = W$, and, by (10),

$$\mathbb{P}(Z_i > r) \leq \mathbb{P}(E_i > r) \leq \Lambda_i(m_i + r), \quad r > 0.$$

Thus $(Z_1, \dots, Z_n) \in \mathcal{Z}_m(X)$. Every feasible allocation with finite upper levels therefore produces an excess-feasible vector with the same total level, proving the reverse inequality. $\square$

Remark 3.3 (What has and has not been simplified). *Theorem 3.2 removes the original full-sum dependence constraint. After choosing m , only the positive excess $W_m = (X - \sum_i m_i)_+$ has to be split. The negative shortfall $(\sum_i m_i - X)_+$ can be assigned arbitrarily among agents and can only decrease their upper tails. The remaining splitting problem is still a genuine mathematical problem; no claim is made here that it always has a closed-form solution.*

4 Attainment and optimal allocations

The preceding proof gives the exact allocation structure attached to any attaining split.

Theorem 4.1 (Attainment criterion and allocation construction). *Let $v := \mathcal{V}(X)$, possibly extended. The level problem (2) is attained if and only if there exist $m^* \in \mathbb{R}^n$ and $(Z_1^*, \dots, Z_n^*) \in \mathcal{Z}_{m^*}(X)$ such that*

$$\sum_{i=1}^n m_i^* = v. \quad (11)$$

Whenever such m^ and Z^* exist, every nonnegative split $D_1, \dots, D_n \in \mathcal{X}$ of the shortfall*

$$\sum_{i=1}^n D_i = (v - X)_+$$

gives an optimal allocation by

$$X_i^* := m_i^* + Z_i^* - D_i, \quad i = 1, \dots, n. \quad (12)$$

Conversely, if $X_1, \dots, X_n$ and finite levels $m_1, \dots, m_n$ attain (2), then the proportional construction in the proof of Theorem 3.2 gives a split in $\mathcal{Z}_m(X)$ with $\sum_i m_i = v$.

Proof. If m^*, Z^* satisfy (11), the allocation (12) sums to X , and the same tail estimate as in the proof of Theorem 3.2 gives $\rho_i(X_i^*) \leq m_i^*$. Thus the total upper level is v , so the level problem is attained.

Conversely, any attaining allocation and attaining finite upper levels produce, by the second half of the proof of Theorem 3.2, a split in $\mathcal{Z}_m(X)$ with the same total level v . $\square$

Remark 4.2 (Not every optimizer is literally canonical). *Formula (12) gives a large canonical family of optimizers. A general optimizer need not be identical to one of these allocations for the same m and Z , because losses below the total level can be redistributed among agents without changing the induced excess split. The correct converse statement is the one in Theorem 4.1: every optimizer with finite attaining upper levels induces an optimal excess split.*

5 A corrected relaxed Kantorovich–Hall criterion

The set $\mathcal{Z}_m(X)$ concerns actual random variables on the given probability space. There is also a distributional, randomized relaxation. This relaxation is useful and has a clean dual, but it is not equivalent to actual splitting on an arbitrary atomless space.

Fix $m \in \mathbb{R}^n$ and let $\mu_m = \text{Law}(W_m)$. For $w \geq 0$, set

$$\Delta_n(w) := \{z = (z_1, \dots, z_n) \in \mathbb{R}_+^n : z_1 + \dots + z_n = w\}.$$

Let $\Gamma_m(X)$ be the set of probability measures π on $\mathbb{R}_+ \times \mathbb{R}_+^n$ such that

$$\pi\{(w, z) : z \in \Delta_n(w)\} = 1, \quad (13)$$

$$\pi \circ w^{-1} = \mu_m, \quad (14)$$

$$\pi\{(w, z) : z_i > r\} \leq b_i^m(r) \quad \text{for all } i = 1, \dots, n \text{ and all } r > 0. \quad (15)$$

Every actual split $Z \in \mathcal{Z}_m(X)$ induces an element of $\Gamma_m(X)$, namely $\text{Law}(W_m, Z_1, \dots, Z_n)$. The converse requires additional randomization.

Let $\mathcal{S}$ be the cone of nonnegative simple tail penalties

$$A(t) = \sum_{k=1}^K c_k \mathbf{1}_{\{t > r_k\}}, \quad K < \infty, \quad c_k \geq 0, \quad r_k > 0.$$

For $A_1, \dots, A_n \in \mathcal{S}$, define

$$\Phi_A(w) := \min_{z \in \Delta_n(w)} \sum_{i=1}^n A_i(z_i), \quad w \geq 0. \quad (16)$$

If $A_i(t) = \sum_k c_{ik} \mathbf{1}_{\{t > r_{ik}\}}$, define

$$B_i^m(A_i) := \sum_k c_{ik} \Lambda_i(m_i + r_{ik}). \quad (17)$$

Theorem 5.1 (Relaxed Kantorovich–Hall criterion). *For fixed $m \in \mathbb{R}^n$, the following are equivalent:*

(a) $\Gamma_m(X) \neq \emptyset$.

(b) For every $A_1, \dots, A_n \in \mathcal{S}$,

$$\mathbb{E}[\Phi_A(W_m)] \leq \sum_{i=1}^n B_i^m(A_i). \quad (18)$$

Consequently, (18) is a necessary condition for $\mathcal{Z}_m(X) \neq \emptyset$. It is also sufficient for $\mathcal{Z}_m(X) \neq \emptyset$ under the richness condition in Proposition 5.2 below.

Proof. Assume first that $\pi \in \Gamma_m(X)$. Since $z \in \Delta_n(w)$ π -a.s.,

$$\Phi_A(w) \leq \sum_{i=1}^n A_i(z_i) \quad \pi\text{-a.s.}$$

Therefore

$$\mathbb{E}[\Phi_A(W_m)] = \int \Phi_A(w) d\pi(w, z) \leq \sum_{i,k} c_{ik} \pi\{z_i > r_{ik}\} \leq \sum_{i,k} c_{ik} \Lambda_i(m_i + r_{ik}),$$

which is (18).

Conversely, suppose (18) holds for all simple penalties. We first record the finite-threshold argument. Fix finitely many pairs (i_j, r_j) , $j = 1, \dots, N$, and write $O_j = \{(w, z) : z_{i_j} > r_j\}$ and $b_j = b_{i_j}^m(r_j)$. Let Π_m be the set of probability measures on $\mathbb{R}_+ \times \mathbb{R}_+^n$ satisfying (13) and (14). This set is nonempty, convex and compact for weak convergence: tightness follows from $0 \leq z_i \leq w$ on the support and from the fixed w -marginal μ_m , and closedness follows from the closedness of the support condition and the fixed marginal condition.

Consider the upper set

$$D := \{y \in \mathbb{R}^N : \exists \pi \in \Pi_m \text{ with } \pi(O_j) \leq y_j \text{ for } j = 1, \dots, N\}.$$

It is convex and upward closed. It is also closed: if $y^\ell \rightarrow y$ and $\pi_\ell(O_j) \leq y_j^\ell$, compactness gives a subsequential weak limit $\pi \in \Pi_m$, and since each O_j is open, the Portmanteau theorem gives

$$\pi(O_j) \leq \liminf_{\ell \rightarrow \infty} \pi_\ell(O_j) \leq y_j.$$

If no $\pi \in \Pi_m$ satisfies all finite constraints $\pi(O_j) \leq b_j$, then $b = (b_1, \dots, b_N) \notin D$. By separating the closed convex upward set D from b , there are coefficients $c_j \geq 0$, not all zero, such that

$$\sum_{j=1}^N c_j b_j < \inf_{\pi \in \Pi_m} \sum_{j=1}^N c_j \pi(O_j). \quad (19)$$

For fixed nonnegative c_j , the right-hand side equals

$$\int \min_{z \in \Delta_n(w)} \sum_{j=1}^N c_j \mathbf{1}_{\{z_{i_j} > r_j\}} d\mu_m(w).$$

Indeed, the integrand is lower semicontinuous in z on the compact set $\Delta_n(w)$, and a measurable minimizer exists by the measurable selection theorem. Inequality (19) is exactly the failure of (18) for the simple penalties obtained by collecting the coefficients c_j by coordinate i_j . Hence (18) implies feasibility for every finite family of threshold constraints.

Now enumerate the positive rationals as $q_1, q_2, \dots$. For each N , the finite-threshold result gives $\pi_N \in \Pi_m$ satisfying

$$\pi_N\{z_i > q_k\} \leq b_i^m(q_k), \quad i = 1, \dots, n, k = 1, \dots, N.$$

By compactness of Π_m , a subsequence converges weakly to some $\pi \in \Pi_m$. For each fixed rational q_k and coordinate i , the set $\{z_i > q_k\}$ is open, so Portmanteau gives

$$\pi\{z_i > q_k\} \leq b_i^m(q_k).$$

Finally, for arbitrary $r > 0$, choose rationals $q_k \downarrow r$. Since both $r \mapsto \pi\{z_i > r\}$ and $r \mapsto b_i^m(r) = \Lambda_i(m_i + r)$ are right-continuous,

$$\pi\{z_i > r\} = \lim_{k \rightarrow \infty} \pi\{z_i > q_k\} \leq \lim_{k \rightarrow \infty} b_i^m(q_k) = b_i^m(r).$$

Thus $\pi \in \Gamma_m(X)$. □

Proposition 5.2 (When the relaxed criterion is exact for actual splits). *Assume the probability space carries a random variable U uniformly distributed on $[0, 1]$ and independent of X . Then, for every $m \in \mathbb{R}^n$,*

$$\Gamma_m(X) \neq \emptyset \implies \mathcal{Z}_m(X) \neq \emptyset.$$

Consequently, under this richness assumption, Theorem 5.1 is an exact dual criterion for $\mathcal{Z}_m(X) \neq \emptyset$.

Proof. Let $\pi \in \Gamma_m(X)$ and disintegrate it as

$$\pi(dw, dz) = \mu_m(dw)K(w, dz).$$

Because the spaces are standard Borel, the randomization lemma gives a Borel map $T : \mathbb{R}_+ \times [0, 1] \rightarrow \mathbb{R}_+^n$ such that $T(w, \cdot)_{\#}\lambda = K(w, \cdot)$ for μ_m -a.e. w , where λ is Lebesgue measure on $[0, 1]$. Set

$$(Z_1, \dots, Z_n) := T(W_m, U).$$

Since U is independent of W_m , the law of (W_m, Z) is π . The support condition (13) gives $\sum_i Z_i = W_m$ a.s., and the tail conditions (15) give (7). Because $0 \leq Z_i \leq W_m$ a.s. and $W_m \in \mathcal{X}$, solidity gives $Z_i \in \mathcal{X}$. $\square$

Corollary 5.3 (Finite-threshold LP test). *Fix m and finitely many thresholds. The corresponding relaxed feasibility problem is an ordinary finite-dimensional linear program whenever W_m has finite support. The dual inequalities in the finite-threshold part of the proof of Theorem 5.1 are exactly the dual inequalities of that LP. If, in addition, the probability space is rich over the level sets of W_m —for instance, if it carries an independent uniform randomizer—then the same finite LP is also exact for actual splittings at those thresholds.*

Proof. If W_m takes values $w_1, \dots, w_J$, with masses p_j , and the threshold set is finite, each simplex $\Delta_n(w_j)$ is partitioned into finitely many cells according to which inequalities $z_i > r$ hold. Assign a nonnegative variable to each nonempty cell and each w_j . The constraints that the cell masses over w_j sum to p_j and that the threshold tail masses are below the prescribed budgets are linear. The separating-hyperplane argument above is precisely finite-dimensional LP duality for this system. The final assertion follows from Proposition 5.2. $\square$

Proposition 5.4 (The relaxed criterion is strictly weaker without richness). *There is a standard atomless probability space, a bounded aggregate loss X , right-continuous functions Λ_1, Λ_2 , and a vector m , such that $\Gamma_m(X) \neq \emptyset$ but $\mathcal{Z}_m(X) = \emptyset$.*

Proof. Let $\Omega = [0, 1]$ with Lebesgue measure, let $X(\omega) = \omega$, take $n = 2$, and set $m_1 = m_2 = 0$. Then $W_m = X$. Define

$$\Lambda_1(x) = \Lambda_2(x) = \begin{cases} 1/2, & x \leq 0, \\ (1-x)/2, & 0 < x < 1, \\ 0, & x \geq 1. \end{cases}$$

These functions are right-continuous and not identically 1. For $0 < r < 1$, the budget is $b_i^m(r) = (1-r)/2$, and for $r \geq 1$ it is 0.

The relaxed problem is feasible: conditionally on $W_m = w$, put $(z_1, z_2) = (w, 0)$ with probability $1/2$ and $(z_1, z_2) = (0, w)$ with probability $1/2$. This defines a measure $\pi \in \Gamma_m(X)$ because

$$\pi\{z_i > r\} = \frac{1}{2}\mathbb{P}(W_m > r) = \frac{1-r}{2}, \quad 0 < r < 1,$$

and the probability is 0 for $r \geq 1$.

Suppose, to the contrary, that an actual split exists. Then $Z_1, Z_2 \geq 0$ and $Z_1 + Z_2 = W_m$ a.s. Letting $r \downarrow 0$ in the tail constraints gives

$$\mathbb{P}(Z_i > 0) \leq \frac{1}{2}, \quad i = 1, 2.$$

But $W_m > 0$ a.s., so at least one of Z_1, Z_2 is positive a.s. Hence

$$1 \leq \mathbb{P}(Z_1 > 0) + \mathbb{P}(Z_2 > 0) \leq 1,$$

and equality forces $\mathbb{P}(Z_1 > 0, Z_2 > 0) = 0$. Therefore $Z_1 = W_m \mathbf{1}_A$ and $Z_2 = W_m \mathbf{1}_{A^c}$ for some measurable $A \subset [0, 1]$.

For every $0 < r < 1$, the two tail constraints give

$$\lambda(A \cap (r, 1]) \leq \frac{1-r}{2}, \quad \lambda(A^c \cap (r, 1]) \leq \frac{1-r}{2}.$$

Since the two left-hand sides sum to $1-r$, both inequalities must be equalities. Thus

$$\lambda(A \cap (r, 1]) = \frac{1-r}{2} \quad \text{for every } 0 < r < 1.$$

The function $r \mapsto \lambda(A \cap (r, 1])$ is absolutely continuous with almost-everywhere derivative $-\mathbf{1}_A(r)$, while $(1-r)/2$ has derivative $-1/2$. This is impossible. Therefore no actual split exists. $\square$

6 Checks and special cases

6.1 Constant Λ : recovery of the classical VaR-plus formula

Let $\Lambda_i \equiv \alpha_i$, where $\alpha_i \in [0, 1]$, and define

$$\text{VaR}_\alpha^+(Y) := \sup\{x \in \mathbb{R} : \mathbb{P}(Y > x) > \alpha\}.$$

Put $\alpha = (\alpha_1 + \dots + \alpha_n) \wedge 1$, with the convention $\text{VaR}_1^+(Y) = -\infty$.

Proposition 6.1. *For constant $\Lambda_i \equiv \alpha_i$,*

$$\left(\bigcap_{i=1}^n \text{VaR}_{\alpha_i}^+ \right)(X) = \text{VaR}_\alpha^+(X) \quad (20)$$

in the upper-level sense of (2).

Proof. Let $s > \text{VaR}_\alpha^+(X)$. Then $\mathbb{P}(X > s) \leq \alpha \leq \sum_i \alpha_i$. Since the space is atomless, the event $\{X > s\}$ can be partitioned into measurable sets $A_1, \dots, A_n$ with $\mathbb{P}(A_i) \leq \alpha_i$. Choose any $m \in \mathbb{R}^n$ with $\sum_i m_i = s$ and set

$$Z_i = (X - s)_+ \mathbf{1}_{A_i}.$$

Then $\sum_i Z_i = (X - s)_+$ and $\mathbb{P}(Z_i > r) \leq \mathbb{P}(A_i) \leq \alpha_i$ for every $r > 0$. By Theorem 3.2, $\mathcal{V}(X) \leq s$. Letting $s \downarrow \text{VaR}_\alpha^+(X)$ gives $\mathcal{V}(X) \leq \text{VaR}_\alpha^+(X)$.

Conversely, suppose m is excess-feasible and let $s = \sum_i m_i$. If $\sum_i \alpha_i \geq 1$, then $\text{VaR}_\alpha^+(X) = \text{VaR}_1^+(X) = -\infty \leq s$. If $\sum_i \alpha_i < 1$, then for any $t > s$ put $r = (t - s)/n > 0$. From $\sum_i Z_i = (X - s)_+$,

$$\{X > t\} \subseteq \bigcup_{i=1}^n \{Z_i > r\}.$$

Therefore

$$\mathbb{P}(X > t) \leq \sum_{i=1}^n \mathbb{P}(Z_i > r) \leq \sum_{i=1}^n \alpha_i = \alpha.$$

No $t > s$ belongs to the defining set for $\text{VaR}_\alpha^+(X)$, so $\text{VaR}_\alpha^+(X) \leq s$. Taking the infimum over feasible m proves the opposite inequality. $\square$

6.2 A quick necessary test

The exact condition $\mathcal{Z}_m(X) \neq \emptyset$ implies many elementary inequalities. For example, if $r_1, \dots, r_n > 0$, then every feasible split satisfies

$$\mathbb{P}\left((X - s(m))_+ > \sum_{i=1}^n r_i\right) \leq \sum_{i=1}^n \Lambda_i(m_i + r_i). \quad (21)$$

Indeed, if $\sum_i Z_i > \sum_i r_i$, then $Z_i > r_i$ for at least one i . This is useful for disproving proposed formulas, but it is not sufficient in general; the full splitting condition or the relaxed Hall family is stronger.

7 Conclusion

The rigorous part of the earlier draft is the excess-splitting idea. After correction, the main theorem gives the exact formula

$$\mathcal{V}(X) = \inf \left\{ \sum_i m_i : (X - \sum_i m_i)_+ \text{ admits an actual tail-capacitated split} \right\}.$$

This is an exact reformulation of the multiple- ΛVaR^+ inf-convolution problem and gives a clean optimal-allocation criterion.

The attempted unrestricted Kantorovich realization in the earlier draft is not valid. A distributional kernel is equivalent to an actual allocation only under additional richness, such as an independent uniform random variable, or in finite situations with enough conditional atomlessness. Without such an assumption the relaxed condition can be strictly weaker, as shown in Proposition 5.4.

Therefore the present note obtains genuine partial progress: a correct exact reduction, a correct allocation theorem, and a correct relaxed LP/dual theory. It does not produce the kind of closed-form expression or universally tractable finite characterization that would fully settle Liu's open problem in its strongest interpretation.

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